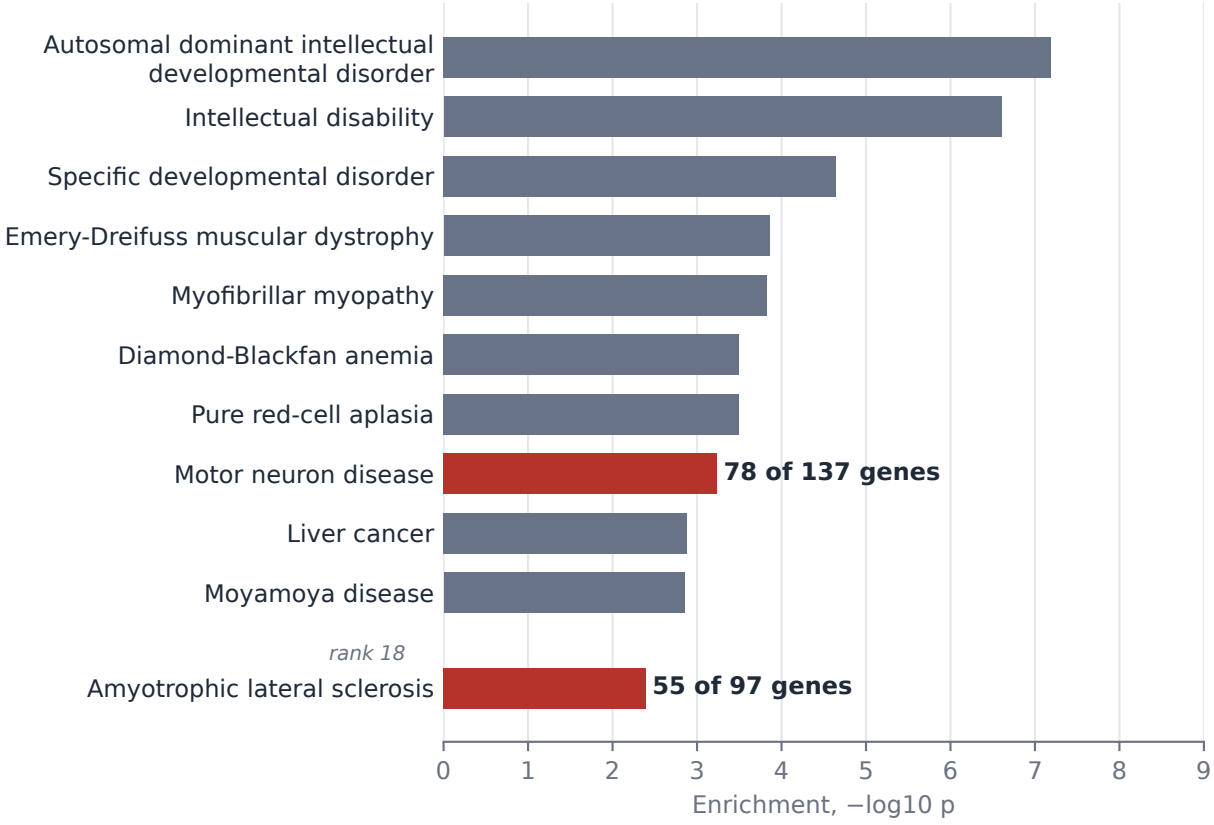
Motor neuron disease ranks 8 of 864 disease terms

Disease Ontology enrichment (DOSE) of 4,687 methylarginine genes against an MS-detectable background. Top 10 terms, plus ALS.



Methylated motor neuron disease genes

TAF15 EWSR1 FUS HNRNPA1 BPTF MATR3 DYNC1H1 SPTAN1 VCP BAG3 UBA1 HSPB1 NEFM YARS1 ANXA11 HSPB8

+ 62 more (ordered by site count)

Fold enrichment 1.33. Background: 5,554 DO-annotated genes detected by MS in human cell lines (PaxDb 5.0) or carrying a methylarginine site, so only proteins that could have been seen are counted.

Motor neuron disease: rank 8 of 864 (p = 0.0006; FDR 0.06). ALS: rank 18 (p = 0.004; FDR 0.19). Top terms are developmental and intellectual-disability terms.

DOSE 4.6 enrichDO (BH). Methylarginine: PTM_pipeline site union (Maron 2021 Table S5, PRoMetheusDB, curated PTM databases; any evidence). Background: PaxDb 5.0 cell-line proteomes (Geiger 2012; Nagaraj 2011; Kulak 2014; Müller 2020).